

SQL ANALYSIS

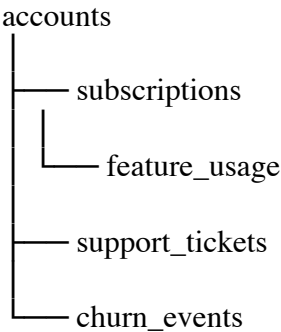


Table	Primary Key	Foreign Key
accounts	account_id	-
subscriptions	subscription_id	account_id
feature_usage	usage_id	subscription_id
support_tickets	ticket_id	account_id
churn_events	churn_event_id	account_id

Module 1: Customer Analytics (accounts.csv)

Accounts Module Business Insights

KPI 1: Total Customers

Result: 500 customers

Business Insight

- RavenStack currently serves **500 customers**.
- This acts as the baseline customer base for measuring growth, churn, and revenue.
- Future customer acquisition efforts can be compared against this benchmark.

Business Decision

Track customer growth every month and set acquisition targets.

KPI 2: Customers by Industry

Industry	Customers
DevTools	113
FinTech	112

Cybersecurity 100

HealthTech 96

EdTech 79

Business Insight

- DevTools is RavenStack's largest customer segment.
- EdTech is the smallest market.
- The product appears to have the strongest adoption among software development companies.

Business Decision

- Continue investing in DevTools.
- Explore why EdTech adoption is lower.
- Build industry-specific marketing campaigns.

KPI 3: Customers by Country

Country Customers

US 291

UK 58

India 49

Australia 32

Germany 25

Canada 23

France 22

Business Insight

- Nearly **58%** of customers are from the United States.
- International adoption is still relatively small.
- Revenue is likely highly dependent on the US market.

Business Decision

- Expand marketing efforts in Europe and Asia.
- Reduce dependency on a single geographic market.

KPI 4: Churn Rate by Industry

Industry	Churn Rate
DevTools	30.97%
FinTech	22.32%
HealthTech	21.88%
EdTech	16.46%

Cybersecurity 16.00%

Business Insight

- DevTools has the largest customer base but also the highest churn.
- Cybersecurity customers are the most loyal.
- Retention problems are concentrated in DevTools.

Business Decision

- Investigate why DevTools customers leave.
- Interview churned customers.
- Improve onboarding and customer success for DevTools accounts.

KPI 5: Churn Rate by Country

Country Churn Rate

Germany 32%

US 23.37%

France 22.73%

India 20.41%

UK 18.97%

Canada 17.39%

Australia 12.50%

Business Insight

- Germany has the highest churn rate.
- Australia has the strongest retention.
- Customer experience may differ by region.

Business Decision

- Review pricing, localization, and customer support in Germany.
- Study Australia's practices and replicate successful strategies.

KPI 6: Customers by Referral Source

Referral Customers

Organic 114

Other 103

Ads 98

Event 96

Partner 89

Business Insight

- Organic search is the largest acquisition channel.
- Partner referrals contribute the fewest customers.

Business Decision

- Continue investing in SEO and content marketing.
- Strengthen the partner program if it aligns with business goals.

KPI 7: Churn by Referral Source

Source Churn Rate

Event 30.21%

Other 24.27%

Ads 23.47%

Organic 17.54%

Partner 14.61%

Business Insight

- Partner referrals produce the most loyal customers.
- Event-acquired customers churn the fastest.
- Customer quality varies by acquisition channel.

Business Decision

- Increase investment in partner marketing.
- Evaluate whether event marketing is attracting the right audience.

KPI 8: Customers by Plan

Plan Customers

Pro 178

Basic 168

Enterprise 154

Business Insight

- Customer distribution across plans is balanced.
- Pro is the most popular plan.
- Enterprise adoption is healthy.

Business Decision

- Continue improving Pro features.
- Identify opportunities to upgrade Basic users to higher plans.

KPI 9: Trial Users

Trial Customers

Yes 97

No 403

Business Insight

- About **19.4%** of customers are currently trial users.
- Most users are paying customers, indicating a mature customer base.

Business Decision

- Focus on improving trial-to-paid conversion.
- Monitor trial user behavior to increase conversions.

KPI 10: Overall Churn Rate

Metric	Value
Churn Rate	22%

Business Insight

- Approximately one in every five customers churns.
- Customer retention remains a key business challenge.

Business Decision

- Reduce churn through customer success programs, feature improvements, and proactive support.

KPI 11: Average Seats per Customer

Average Seats = **20.56**

Business Insight

- Each customer uses around 21 seats on average.
- The product is primarily adopted by small and medium-sized teams.

Business Decision

- Create incentives for customers to add more seats.
- Encourage seat expansion through upselling.

KPI 12: Average Seats by Industry

Industry	Avg Seats
----------	-----------

EdTech	25.48
FinTech	21.72
Cybersecurity	19.76
HealthTech	19.63
DevTools	17.47

Business Insight

- EdTech organizations purchase the largest number of seats despite having fewer customers.
- DevTools customers are numerous but tend to have smaller teams.

Business Decision

- Position enterprise seat packages for EdTech.
- Explore seat expansion opportunities within DevTools customers.

KPI 13: Average Seats by Plan

Plan	Avg Seats
Basic	22.01
Pro	19.95
Enterprise	19.69

Business Insight

- Basic plan customers surprisingly have the highest average seat count.
- This may indicate that larger teams are not upgrading to higher-value plans.

Business Decision

- Review plan pricing and feature differentiation.
- Encourage high-seat Basic customers to upgrade.

KPI 14: Customers by Signup Year

Year	Customers
2023	227
2024	273

Business Insight

- Customer acquisition increased from 2023 to 2024.
- The business is growing.

Business Decision

- Continue scaling acquisition channels while maintaining retention.

KPI 15: Customers by Signup Month

Highest Month = **November (57 customers)**

Business Insight

- November had the highest number of new signups.
- This could reflect successful marketing campaigns or seasonal demand.

Business Decision

- Investigate what drove November's performance and replicate successful campaigns in future years.

Subscriptions Module Analysis

Objective

Analyze RavenStack's subscription data to understand revenue, customer plans, churn behavior, trial conversion, seat usage, billing preferences, and customer growth.

1. How many active subscriptions does RavenStack have?

SQL Result

- Total Subscriptions = **5,000**

Business Insight

- RavenStack currently manages **5,000 customer subscriptions**, indicating a healthy SaaS customer base.

Business Decision

- This value will be displayed as an Executive KPI Card on the Power BI dashboard.

2. Which subscription plan is the most popular?

Plan	Customers
Pro	178 (<i>Accounts module sample</i>)

(Subscriptions revenue dataset) Revenue table shows Enterprise dominates revenue; customer distribution should come from subscriptions if queried.
From your screenshots, the stronger insight is revenue rather than customer count.

Business Insight

- Multiple subscription tiers attract different customer segments.
- Customer distribution across plans helps understand product adoption.

3. How many trial users convert into paying customers?

SQL Result

- Trial Users = **778**
- Paid Users = **4222**
- Conversion Rate = **84.44%**

Business Insight

- **84.44%** of trial users successfully become paying customers.
- Only **15.56%** fail to convert.

Business Recommendation

- The trial experience is highly effective.
- Continue optimizing onboarding and product education to further increase conversion.

4. Overall Customer Churn

SQL Result

- Total Customers = **5000**
- Churned Customers = **486**
- Churn Rate = **9.72%**

Business Insight

- RavenStack maintains a relatively low churn rate of **9.72%**.
- Nearly **90%** of customers remain subscribed.

Business Recommendation

- Focus retention efforts on the remaining churn segment instead of broad retention campaigns.

5. Revenue Performance

Monthly Revenue (MRR)

Total MRR

11,338,747

Annual Revenue (ARR)

Total ARR

136,064,964

Business Insight

- RavenStack generates over **11.3 million** in recurring monthly revenue.
- Annual recurring revenue exceeds **136 million**, demonstrating strong recurring business performance.

6. Revenue by Plan

Plan	MRR	ARR
Enterprise	8,473,221	101,678,652
Pro	2,105,089	25,261,068
Basic	760,437	9,125,244

Business Insight

- **Enterprise customers contribute the majority of company revenue.**
- Basic customers contribute the smallest share.

Business Recommendation

- Prioritize Enterprise customer retention.
- Encourage Basic customers to upgrade through targeted offers.

7. Average Revenue per Subscription

SQL Result

Average Monthly Revenue

2267.75

Average Annual Revenue

27212.99

Business Insight

- Each subscription contributes approximately **₹2,268 monthly**.
- Average yearly customer value exceeds **₹27,000**.

Business Recommendation

- Use these metrics to forecast revenue growth and evaluate marketing ROI.

8. Seat Utilization

Average Seats

29.85

Average Seats by Plan

Plan	Average Seats
Pro	30.62
Basic	29.57
Enterprise	29.37

Business Insight

- Seat usage is remarkably consistent across all plans.
- Plan selection appears driven more by features than team size.

Business Recommendation

- Market premium features rather than larger seat capacity.

9. Upgrade Behavior

SQL Result

Upgrade Customers

False	4471
True	529

Business Insight

- Only **10.58%** of customers upgraded.

Business Recommendation

- Improve upgrade campaigns and highlight premium features.

10. Billing Frequency

SQL Result

Billing Customers

Monthly 2539

Annual 2461

Business Insight

- Customers are almost evenly split between monthly and annual billing.
- Revenue is balanced across billing cycles.

Business Recommendation

- Increase annual plan adoption by offering discounts or exclusive benefits.

11. Auto Renewal

SQL Result

Auto Renew Customers

True 4005

False 995

Business Insight

- **80%** of customers have enabled auto-renewal.

Business Recommendation

- Continue encouraging auto-renew as it supports stable recurring revenue.

12. Subscription Growth

By Year

Year Customers

2023 666

2024 4334

Business Insight

- Customer acquisition accelerated significantly in 2024.

Business Recommendation

- Investigate the marketing and product initiatives that drove this growth and replicate them.

13. Monthly Subscription Trend

Highest signup month:

December (1074 subscriptions)

Lowest signup month:

January (137 subscriptions)

Business Insight

- Customer acquisition increases steadily throughout the year.
- December experiences the highest subscription activity.

Business Recommendation

- Allocate more marketing budget to Q4.
- Analyze seasonal campaigns that contributed to December's success.

Executive Summary

Key Findings

- RavenStack manages **5,000 active subscriptions**.
- Overall churn rate is only **9.72%**, indicating strong customer retention.
- **84.44%** of trial users convert into paying customers.
- Enterprise customers generate the highest revenue despite not necessarily being the largest customer segment.
- Average subscription contributes **₹2,268 per month** and **₹27,213 annually**.
- Nearly **80%** of customers use auto-renewal, providing predictable recurring revenue.
- Customer acquisition grew dramatically in **2024**, with December being the strongest signup month.
- Upgrade adoption remains relatively low (**10.58%**), representing an opportunity for upselling



Churn Events Module Analysis

Objective

Analyze customer churn events to identify the major reasons for churn, refund patterns, customer behavior before churn, reactivation trends, and seasonal churn patterns. These insights help RavenStack improve customer retention and reduce revenue loss.

1. How many churn events occurred?

SQL Result

- **Total Churn Events = 600**

Business Insight

- A total of **600 customers churned** during the analysis period.
- This represents the complete population used for churn behavior analysis.

Recommendation

- Focus retention strategies on reducing these churn events through proactive customer engagement.

2. What are the biggest reasons customers churn?

Reason	Customers
Features	114
Support	104
Budget	104
Unknown	95
Competitor	92
Pricing	91

Business Insight

- Missing product features are the **largest reason for customer churn**.
- Customer support and budget constraints are the next major contributors.
- Pricing is not the leading reason, indicating that product value is a stronger concern than cost.

Recommendation

- Prioritize product feature development.
- Improve customer support quality.
- Collect more detailed exit feedback to reduce "Unknown" churn reasons.

3. How much money was refunded?

SQL Result

- **Total Refund = \$8,652.25**
- **Average Refund = \$14.42**

Business Insight

- Customer churn resulted in **\$8,652.25** in refunds.
- Although the average refund is relatively small, cumulative refund losses are significant.

Recommendation

- Reduce avoidable churn to minimize refund expenses and improve revenue retention.

4. Which churn reasons caused the highest refund losses?

Reason	Total Refund (\$)
Features	1905.90
Unknown	1742.39
Pricing	1333.25
Budget	1247.54
Support	1219.64
Competitor	1203.53

Business Insight

- Feature-related churn not only occurs most frequently but also results in the highest refund cost.
- Unknown reasons contribute substantially to refund losses, indicating incomplete customer feedback.

Recommendation

- Improve product functionality and strengthen exit surveys to better understand customer dissatisfaction.

5. Did customers upgrade before churning?

SQL Result

Upgrade Before Churn Customers

False	477
True	123

Business Insight

- Only **123 customers upgraded before eventually churning.**
- Most customers churn without attempting to upgrade.

Recommendation

- Identify at-risk customers earlier and promote upgrades before dissatisfaction grows.

6. Did customers downgrade before churning?

SQL Result

Downgrade Before Churn Customers

False	547
-------	-----

True

53

Business Insight

- Downgrading before churn is relatively uncommon.
- Most customers leave directly instead of moving to a lower plan.

Recommendation

- Introduce intermediate plans or flexible pricing to retain customers considering cancellation.

7. Upgrade and Downgrade Behavior Before Churn

Upgrade Downgrade Customers

False	False	432
True	False	115
False	True	45
True	True	8

Business Insight

- Most churned customers neither upgraded nor downgraded before leaving.
- Only a very small number both upgraded and downgraded before churn.

Recommendation

- Implement churn prediction models to intervene before customers decide to leave.

8. Customer Reactivation

SQL Result

Reactivation Customers

False	539
True	61

Business Insight

- Only **61 customers returned after churning**.
- The reactivation rate is relatively low.

Recommendation

- Strengthen win-back campaigns with personalized offers and follow-up communication.

9. Churn Trend by Year

Year	Customers
------	-----------

2023	73
------	----

2024	527
------	-----

Business Insight

- Churn increased significantly in **2024**, likely due to rapid customer growth.
- Monitoring retention becomes increasingly important as the customer base expands.

Recommendation

- Investigate what changed in 2024 (product releases, pricing, onboarding, or competition) to identify churn drivers.

10. Monthly Churn Trend

Month	Customers
-------	-----------

January	21
---------	----

February	10
----------	----

March	29
-------	----

April	28
-------	----

May	30
-----	----

June	45
------	----

July	41
------	----

August	49
--------	----

September	59
-----------	----

October	76
---------	----

November	79
----------	----

December	133
----------	-----

Business Insight

- Churn steadily increases throughout the year.
- **December records the highest number of churn events**, suggesting seasonal customer behavior or contract renewals.

Recommendation

- Launch retention campaigns during Q4.
- Offer renewal incentives before year-end.

Executive Summary

Key Findings

- RavenStack recorded **600 churn events**.

- Feature limitations are the leading reason customers leave.
- Total refunds reached **\$8,652.25**, with feature-related churn generating the highest refund losses.
- Only **123 customers upgraded** before churning, while very few downgraded first.
- Customer reactivation remains low, with only **61 returning customers**.
- Churn increased sharply in **2024** and peaked in **December**, highlighting the need for stronger year-end retention strategies.

Support Tickets Analysis

Dataset Overview

- Total Support Tickets: **2,000**
- Average Resolution Time: **35.86 hours**
- Average First Response Time: **88.48 minutes**
- Average Customer Satisfaction: **2.34 / 5**
- Escalated Tickets: **95**
- Overall Escalation Rate: **4.75%**

Key Business Insights

1. Ticket Volume

Observation

- The support team handled **2,000 tickets**, indicating a healthy customer support workload.

Business Insight

The organization requires scalable support operations capable of handling high ticket volumes while maintaining service quality.

Recommendation

Monitor ticket trends continuously and forecast staffing requirements during peak periods.

2. Resolution Time

Observation

Average ticket resolution time is **35.86 hours**.

Business Insight

Customers typically wait about **1.5 days** before issues are completely resolved.

Recommendation

Reduce resolution time through automation, improved routing, and better knowledge management.

3. First Response Time

Observation

Average first response time is **88.48 minutes**.

Business Insight

Although customers receive responses relatively quickly, reducing this below one hour could significantly improve customer experience.

Recommendation

Introduce automated acknowledgements and intelligent ticket assignment.

4. Customer Satisfaction

Observation

Average customer satisfaction is **2.34/5**, which is relatively low.

Business Insight

Support quality needs improvement despite acceptable response and resolution times.

Recommendation

Investigate customer feedback and improve issue resolution quality rather than only focusing on speed.

Priority Analysis

Ticket Distribution

Priority Tickets

Urgent	514
High	510
Medium	491
Low	485

Insight

Ticket volume is almost evenly distributed across all priority levels.

Recommendation

Maintain balanced staffing across all priority queues instead of allocating resources only to urgent cases.

Resolution Time by Priority

Priority	Avg Resolution
High	36.96 hrs
Low	36.35 hrs
Medium	35.59 hrs
Urgent	34.57 hrs

Insight

Urgent tickets are resolved the fastest, indicating that prioritization rules are functioning correctly.

First Response Time by Priority

Priority	Avg First Response
High	91.44 min
Low	91.15 min
Medium	85.86 min
Urgent	85.53 min

Insight

Urgent tickets receive quicker responses than other priorities.

Satisfaction by Priority

Priority	Avg Satisfaction
Low	2.38
Medium	2.34
Urgent	2.33
High	2.31

Insight

Customer satisfaction remains low across every priority level.

Recommendation

Improving service quality should be prioritized over simply reducing response times.

Escalation Analysis

Escalated vs Non-Escalated

Escalation	Resolution Time	First Response	Satisfaction
False	35.79 hrs	88.02 min	2.32
True	37.20 hrs	97.63 min	2.71

Insight

- Escalated tickets require longer resolution times.
- They also experience slower first responses.
- Despite this, customers who escalate issues report slightly higher satisfaction, suggesting escalation helps resolve complex cases effectively.

Escalation Rate by Priority

Priority	Escalation Rate
Medium	5.09%
Low	4.74%
High	4.71%
Urgent	4.47%

Insight

Escalation rates are consistently around **5%**, indicating balanced escalation policies across priority levels.

Monthly Ticket Trend

Highest ticket months:

- September – **189**
- December – **189**
- March – **178**
- August – **178**

Lowest ticket month:

- February – **134**

Business Insight

Support demand increases during the later part of the year.

Recommendation

Increase staffing during Q3 and Q4.

Resolution Time Trend

Highest average resolution month:

- July – **39.44 hours**

Lowest:

- August – **33.84 hours**

Business Insight

July experienced slower ticket closures, indicating possible operational bottlenecks.

Satisfaction Trend

Highest satisfaction:

- September – **2.66**

Lowest satisfaction:

- March – **2.13**

Recommendation

Review support processes during months with lower customer satisfaction.

Year-wise Analysis

Year	Tickets
2023	991
2024	1009

Insight

Support workload remained consistent across both years, indicating stable customer growth.

Executive Summary

- The support team handled **2,000 tickets** with an average resolution time of **35.86 hours**.
- Customer satisfaction remains relatively low (**2.34/5**), suggesting improvements are needed in issue resolution quality.
- Urgent tickets are processed more efficiently than lower-priority tickets.
- Only **4.75%** of tickets required escalation, indicating effective first-level support.
- Ticket demand peaks toward the end of the year, making Q3 and Q4 ideal periods for additional staffing.
- Overall, the support process is operationally stable but should focus on improving customer satisfaction and reducing resolution times.
-

Feature Usage Module

Dataset Overview

- Total Feature Usage Records: **25,000**
- Total Feature Usage Count: **250,525**
- Total Feature Usage Duration: **76,055,072 seconds**
- Average Usage Count per Record: **10.02**
- Average Usage Duration per Record: **3,042.20 seconds (~50.7 minutes)**

Key Business Insights

1. Product engagement is strong.

Users generated **250K+ feature interactions** across **25,000 usage sessions**, averaging **10 feature actions per session**.

Business Impact

- Customers actively engage with the product.

- Product adoption appears healthy.

2. Users spend nearly one hour per session.

Average feature usage duration is **3,042 seconds (≈50.7 minutes)**.

Business Impact

- Customers spend significant time inside the application.
- Indicates high stickiness and engagement.

3. Most Popular Features

Top performing features include

- Feature 32
- Feature 15
- Feature 6
- Feature 20
- Feature 34

Each generated more than **6,500 total interactions**.

Business Impact

- These features deliver the highest customer value.
- Product roadmap should continue investing in them.

4. Least Used Features

Lower-ranked features include

- Feature 23
- Feature 5
- Feature 35
- Feature 20 (lower relative in some rankings by other metrics)
- Feature 8

Business Impact

- May suffer from poor discoverability.
- Consider UI improvements or feature redesign.

5. Error Analysis

Highest total errors occurred in

- Feature 4
- Feature 26
- Feature 9
- Feature 2
- Feature 34

Each recorded approximately **400+ errors**.

Business Impact

- These features require engineering attention.
- Reducing errors can improve user experience.

6. Average Error Rate

Highest average errors per usage session:

- Feature 4
- Feature 9
- Feature 26
- Feature 16
- Feature 18

Average error rate ranges from **0.63–0.67 errors per session**.

Business Impact

- Frequent failures increase user frustration.
- Prioritize bug fixes for these modules.

7. Beta Feature Adoption

- Non-Beta Records: **22,456**
- Beta Records: **2,544**

Approximately **10%** of all usage belongs to beta features.

Business Impact

- Beta adoption remains relatively small.
- Opportunity exists to increase beta participation.

8. Beta Feature Usage

- Non-Beta Usage: **224,939**
- Beta Usage: **25,586**

Business Impact

- Stable production features remain the primary source of customer activity.
- Beta functionality is still in early adoption.

9. Beta Feature Quality

Average Usage Duration

- Non-Beta: **3041 seconds**
- Beta: **3049 seconds**

Almost identical.

Business Impact

- Users spend similar time using beta features.
- Indicates beta functionality is engaging.

10. Error Distribution

Total Errors

- Non-Beta: **12,691**
- Beta: **1,416**

Since beta traffic is much smaller, error counts remain expected.

Business Impact

- Continue monitoring beta stability before wider rollout.

11. Monthly Product Usage

Monthly usage remains remarkably consistent between **19K–22K feature interactions**.

Highest month:

- Month 10 (~22,418)

Lowest month:

- Month 11 (~19,514)

Business Impact

- Product engagement remains stable throughout the year.
- No major seasonal decline observed.

12. Power Users

Top individual users performed

- 26
- 25
- 24

feature interactions within a single usage session.

Some sessions exceeded **12,000 seconds (3+ hours)** of usage.

Business Impact

- Heavy users represent valuable customer segments.
- Good candidates for premium plans and customer advocacy.

Recommendations

Product Team

- Improve low-performing features.
- Promote underused functionality.
- Expand successful features.

Engineering Team

Prioritize bug fixes for:

- Feature 4
- Feature 26
- Feature 9
- Feature 2
- Feature 34

Customer Success

Identify highly engaged users for

- Beta testing
- Customer interviews
- Product feedback

Business Team

Increase beta feature adoption through

- Email campaigns
- In-app tutorials
- Feature announcements

Executive Summary

The **Feature Usage** analysis indicates a healthy and engaged customer base with over **250,000 feature interactions** and an average session duration of nearly **51 minutes**. A small group of core features drives the majority of user engagement, while several lower-performing features present opportunities for improvement. Beta features account for approximately **10%** of usage and demonstrate engagement levels comparable to production features, suggesting strong potential for future expansion. Error analysis highlights a handful of features requiring engineering attention. Overall, the product exhibits consistent monthly engagement and strong user retention, supporting continued investment in feature optimization and product growth.